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A. (30pts) We have a liquid feed that is 40.0 mol% methane (C1), 20.0 mol% n-butane (C4), 15.0 mol% n-pentane (C5) and unknown mol% of n-hexane (C6). Feed rate (F) is 100 kmol/h and fraction feed vaporized (V/F) is 0.7. Feed pressure is 100.0 psia. The flash drum operates at 250 kPa and 10 °C.

- 1) (5pts) Sketch the process.
- 2) (5 pts) Find mole fraction (z_4) of n-hexane in the feed and flow rate of vapor stream (V) and liquid stream (L).
- 3) (15 pts) Find V/F use Rachford-Rice Equations. $K_{C1} = 50$, $K_{C4} = 0.6$, $K_{C5} = 0.17$, and $K_{C6} = 0.05$. Use $(V/F)_1 = 0.45$ as 1st guess and find V/F value when $f(V/f)$ converges to less than 0.01.

$$f\left[\frac{V}{F}\right] = \sum_{i=1}^c \frac{(K_i - 1)z_i}{1 + (K_i - 1)\left(\frac{V}{F}\right)} = 0$$

$$\left[\frac{V}{F}\right]_{k+1} = \left[\frac{V}{F}\right]_k - \frac{f_k}{\left[\frac{df_k}{d(V/F)}\right]}$$

$$\frac{df_k}{d(V/F)} = - \sum_{i=1}^c \frac{(K_i - 1)^2 z_i}{\left[1 + (K_i - 1)\left(\frac{V}{F}\right)\right]^2}$$

- 4) (5 pts) Find x_i , y_i , V, and L

$$x_i = \frac{z_i}{1 + (K_i - 1)\frac{V}{F}} \quad i=1, C \quad y_i = K_i x_i = \frac{K_i z_i}{1 + (K_i - 1)\frac{V}{F}} \quad i=1, C$$

- 5) (5 pts) In the class, we practiced the double trial and error (simultaneous multicomponent convergence) procedures when T_{drum} is unknown. Make your own flowsheet to find T_{drum} , V/F, x_i , y_i , L, and V

B. (70 pts) We are separating acetone and ethanol in a stripping column. The column has two feeds and operates at a pressure of 1.0 atm. Feed 1 is a saturated liquid and is fed into the top of column (no condenser). Flow rate of F1 is 150.0 kmol/hr with 60.0 mol% acetone. Feed 2 is a 10.0 mol% acetone and two-phase mixture with that is 80% vapor. Flow rate of F2 = 50.0 kmol/hr. Assume CMO is valid. Bottoms product is 4.0 mol% acetone and the column has a partial reboiler with bottom up. Boil-up ratio ($\bar{V}/B$) is 1.5.

- 1) (5 pts) Draw a schematic diagram of the distillation column and label all known variables (use V & L for enriching section and $\bar{L}$ & $\bar{V}$ for stripping section)
- 2) (5 pts) There are total three balance envelopes required for this question (enriching & Feed 1 section, stripping section, and Feed2). Draw separately from (a) and write down mass balance for each section.
- 3) (5 pts) Set up overall mass balance and component mass balance
- 4) (10 pts) Find L, $\bar{L}$, B, $\bar{V}$, V, and D

5) (10 pts) The equation for the feed line is:

$$y = \frac{q_i}{z_i} x + \frac{1 - q_i}{z_i}$$

$$q_i = \frac{L_{Fi}}{F_i}$$

Find q_1 , q_2 , coordinates on $y=x$ line, and slopes of two feed lines (F_1 & F_2), and y -intercept of F_2 .

6) (5 pts) Find mole composition of ethanol in distillate stream (y_D)

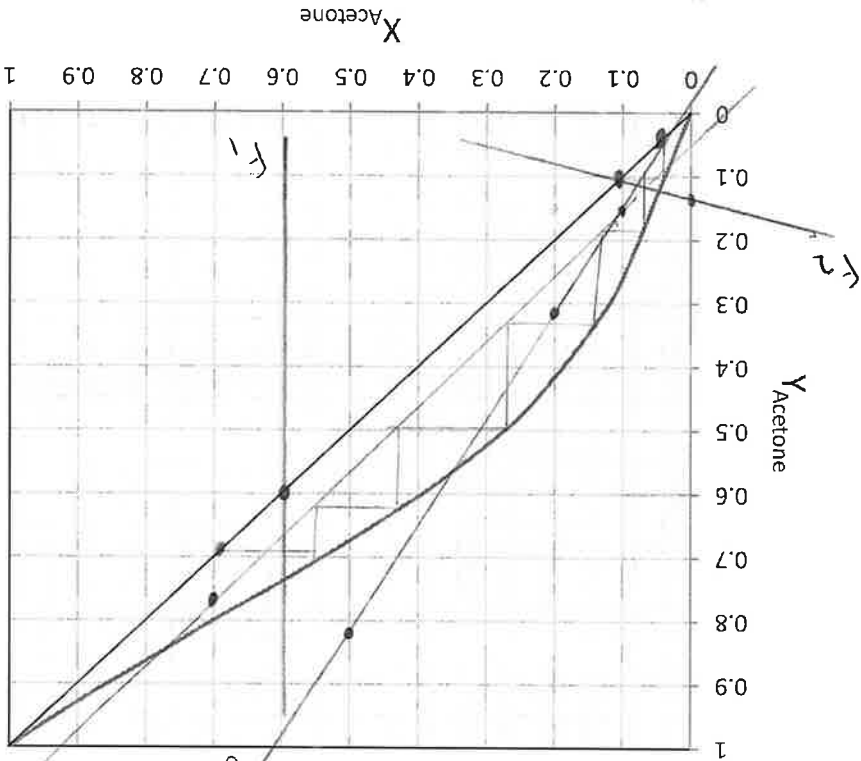
7) (5 pts) Set up component mass balance equation for the top & feed 1 section using mass balance from (b). Then, derive operating equation for the top section.

8) (5 pts) Find the slope, y -intercept, and the intersection point of Feed 1 and top operating line.

9) (5 pts) Find slope and coordinate on $y=x$ curve of the bottom operating line.

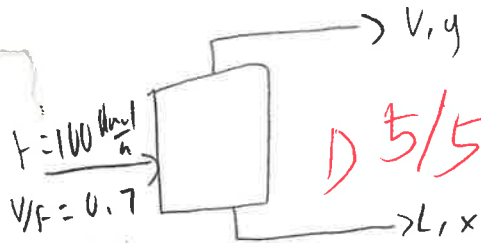
$$y = \left(\frac{L}{D}\right)x - \left(\frac{L}{D} - 1\right)x_B$$

10) (10 pts) Draw the top operating, bottom operating, two feeds, and middle operating lines



11) (5 pts) Find the number of stages and the optimum locations of feed 1 and feed 2 stages. Step off stage from the bottom.

A)



$$2) I = z_{c1} + z_{c4} + z_{c5} + z_{c6}$$

$$I = 0.4 + 0.2 + 0.15 + z_{c6} \rightarrow 5/5$$

$$z_{c6} = 0.25 = z_{c4}$$

$$\frac{V}{F} = 0.7 \rightarrow V = 0.7F = 70 \frac{\text{kmol}}{\text{h}}$$

$$F = L + V \rightarrow L = F - V = 30 \frac{\text{kmol}}{\text{h}}$$

$$z_{c1} = 0.4 \quad k_{c1} = 50$$

$$z_{c4} = 0.2 \quad k_{c4} = 0.6$$

$$z_{c5} = 0.15 \quad k_{c5} = 0.17$$

$$z_{c6} = \text{---} \quad k_{c6} = 0.05 \rightarrow 0.6$$

$$3) f\left(\frac{V}{F}\right) = \frac{(50-1)0.4}{1+(50-1)\frac{V}{F}} + \frac{(0.6-1)0.2}{1+(0.6-1)\frac{V}{F}} + \frac{(0.17-1)0.15}{1+(0.17-1)\frac{V}{F}} + \frac{(0.05-1)0.25}{1+(0.05-1)\frac{V}{F}}$$

$$\frac{df_k}{d(V/F)} = \frac{(50-1)^2 0.4}{[1+(50-1)\frac{V}{F}]^2} + \frac{(0.6-1)^2 0.2}{[1+(0.6-1)\frac{V}{F}]^2} + \frac{(0.17-1)^2 0.15}{[1+(0.17-1)\frac{V}{F}]^2} + \frac{(0.05-1)^2 0.25}{[1+(0.05-1)\frac{V}{F}]^2}$$

$$\left(\frac{V}{F}\right)_1 = 0.45 \quad \frac{df_k}{d(V/F)} = 1.8076 + 0.047591 + 0.26327 + 0.68839 = 2.806851$$

$$f(0.45) = 0.85032 - 0.097561 - 0.198723 - 0.414847 = 0.139189 > 0.01$$

$$\left(\frac{V}{F}\right)_2 = 0.45 - \frac{0.139189}{2.806851} = 0.40$$

$$\left(\frac{V}{F}\right)_2 = 0.40 \quad \frac{df_k}{d(V/F)} = 2.26317 + 0.045351 + 0.231576 + 0.586953 = 3.12705$$

$$f(0.4) = 0.951456 - 0.0952380 - 0.186377 - 0.38366451 = 0.485776$$

$$\left(\frac{V}{F}\right)_1 = 0.45 = \frac{df_k}{d(V/F)} = -2.59 \quad f(0.45) = 0.8503254 - 0.09756097 - 0.1987230 - 0.414847 = 0.139189 > 0.01$$

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$$\left(\frac{V}{F}\right)_2 = 0.45 + \frac{0.139189}{2.59} = 0.396$$

0.50 ← V/F

$$\left(\frac{V}{F}\right)_2 = 0.396 \quad \frac{df_k}{d(V/F)} = 3.431535419 \quad f(0.396) = 1.02719983 - 0.0950870342 - 0.1854555 - 0.603077 = 0.1436$$

$f(V/F=0.50) = 0.0061 < 0.01$
 $\therefore V/F = 0.50$

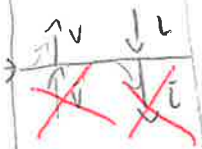
5)



B) ... (sat 1.1) : $q = 1$

$$F_i = L$$

1) $F_i = 50$
 $z_i = 0.16$



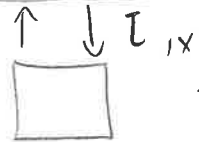
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$$F_2 = 50 \frac{\text{kgm}}{\text{s}}$$

$$z_2 = 0.1$$

$$F = 0.8$$

$$q = 0.2$$

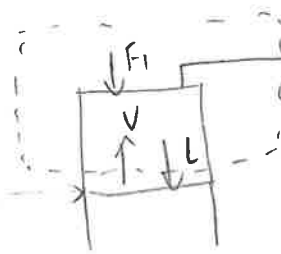


$$\bar{V}/B = 1.5$$

$$B \times B = 0.04$$

Feed 2 ($f = 0.8 \rightarrow q = 1 - f = 0.2$) (MO: $F_i = L$)

2)



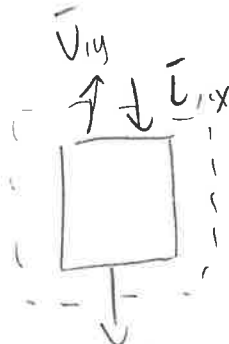
$$V + F_i = D + L$$

$$V = L + D \rightarrow D = V - L$$

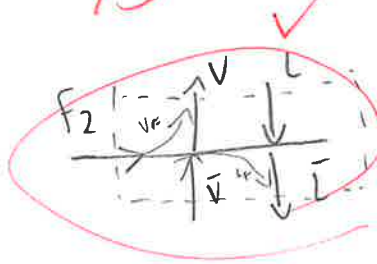
$$V_y = Lx + D \times D$$

$$y = \frac{L}{V}x + \frac{D}{V} \times D$$

$$y = \frac{L}{V}x + (1 - \frac{L}{V}) \times D$$



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$$\bar{L} = \bar{V} + B \rightarrow B = \frac{\bar{L} - \bar{V}}{\bar{V}}$$

$$\bar{L}x = \bar{V}y + B \times B$$

$$y = \frac{\bar{L}}{\bar{V}}x - \frac{B}{\bar{V}} \times B$$

$$y = \frac{\bar{L}}{\bar{V}}x - (\frac{\bar{L}}{\bar{V}} - 1) \times B$$

$$\bar{L} + V = F_2 + L + \bar{V}$$

$$\bar{L}x + Vy = F_2 + Lx + V$$

$$q_2 = \frac{LF_2}{F_2} = \frac{\bar{L} - L}{\bar{F}_2}$$

3)



$$F_1 + F_2 = D + B$$

$$F_1 z_1 + F_2 z_2 = D y_D + B \times B$$

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4) $q_2 = \frac{LF_2}{F_2} = \frac{\bar{L} - L}{F_2}$ $0.2 = \frac{\bar{L} - 150}{50}$ $F_1 = L = 150 \frac{\text{kmol/hr}}$ $f = 0.8$
 $q = 1 - f = 1 - 0.8 = 0.2$

* use this $10 = \bar{L} - 150 \rightarrow \bar{L} = 160 \frac{\text{kmol/hr}}$ ✓

$f_2 = \frac{VF_2}{F_2} = \frac{V - \bar{V}}{F_2} = \frac{V - 96}{50} = 0.8$ $\bar{L} = \bar{V} + B$ $\frac{\bar{V}}{B} = 1.5 \rightarrow \bar{V} = 1.5B$
 $\bar{L} = 1.5B + B = 2.5B \rightarrow B = \frac{160}{2.5} = 64 \frac{\text{kmol/hr}}$

$160 = \bar{V} + 64 \rightarrow \bar{V} = 96 \frac{\text{kmol/hr}}$ ✓
 $V = 136 \frac{\text{kmol/hr}}$ ✓
 $F_1 + F_2 = D + B \rightarrow D = F_1 + F_2 - B = 150 + 50 - 64 = 136 \frac{\text{kmol/hr}}$ ✓

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4) Ho.p. op ~~can: $y = \frac{L}{V}x + (1 - \frac{L}{V})x_D$~~ $* F_1 = L$ (M0) $L = 150 \frac{\text{mol}}{\text{h}}$
 $F_2: 0.2 = \frac{\bar{L} - L}{F_2} = \frac{\bar{L} - 150}{50} \quad \bar{L} = 160 \frac{\text{mol}}{\text{h}}$ $* f = 0.8 \rightarrow q = 1 - f = 0.2$
 $1 \cdot L_{F_2} =$ vapor liquid

~~$\bar{L} = \bar{V} + B$ $\frac{\bar{V}}{B} = 1.5 \rightarrow \bar{V} = 1.5B \rightarrow 160 = 1.5B + B \rightarrow B = 64 \frac{\text{mol}}{\text{h}}$
 $160 = \bar{V} + 64 \rightarrow \bar{V} = 96 \frac{\text{mol}}{\text{h}}$
 $0.8 = \frac{V_{F_1}}{F_2} = \frac{V - \bar{V}}{F_2} = \frac{V - 96}{50} \rightarrow V = 136 \frac{\text{mol}}{\text{h}}$
 $V = L + D \rightarrow 136 = 150 + D \rightarrow D = -14$~~

5) $q_1 = 1$ (sat liq) $y = \frac{q_1}{q_1 - 1}x + \frac{z_1}{1 - q_1}$ $\rightarrow y = 0.6$
 $(0.6, 0.6)$ on $y = x$ $\text{slope: } \infty = \frac{1}{0}$ ✓

$q_2 = 1 - f = 0.2, z_2 = 0.1$ $(0.1, 0.1)$ on $y = x$

$y = \frac{q_2}{q_2 - 1}x + \frac{z_2}{1 - q_2}$

$y = -0.25x + 0.125$ ✓
 $y = -1/4x + 1/8$

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$$6) y_D = \frac{F_1 z_1 + F_2 z_2 - B x_B}{D} = \frac{(156)(0.6) + (50)(0.1) - (64)(0.64)}{136} = 0.68$$

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7) top op eqn: $V = L + D$, $V y = L x + D y_D \rightarrow y = \frac{L}{V} x + \frac{D}{V} y_D$

$$D = V - L \rightarrow \frac{D}{V} = 1 - \frac{L}{V} \quad y = \frac{L}{V} x + (1 - \frac{L}{V}) y_D$$

3/5 feed 1: $q_1 = 1$ (sat liq) $y = \frac{q_1}{q_1 - 1} x + \frac{z_1}{1 - q_1}$ ($z_1 = 0.6$, $q_1 = 1$)

Top ~~plate~~ op: $y = \frac{L}{V} x + \frac{D y_D - F_1 z_1}{V} = 1.1x + 0.068$

8) top op eqn: slope: $\frac{L}{V} = \frac{156}{136} = 1.1$ y-inter: $(1 - \frac{L}{V}) y_D = (1 - \frac{156}{136})(0.68)$

$$0.068 = -0.07$$

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$$y = 1.1x - 0.07$$

(0.6, 0.68) Intersection point.

feed 1: $y = \frac{1}{1-1} x + \frac{0.6}{0} = 0.6$

$$0.6 = 1.1x - 0.07$$

(0.604, 0.6) [intersection]

9) (0.04, 0.04) &